1.Write an ECP to Implement Password Authentication system using UART interrupt, password is correct/incorrect repeat the program in infinite loop.

Example:

Enter the password: ***** (press enter key)

Result : correct/incorrect password

2.Write an ECP, to print temperature sensor output on the serial terminal every time when the enter key is pressed on the serial terminal.

Connections- MCP9700 Temperature Sensor connected to P0.28 (Ain1) pin.

Eg: On the serial terminal

Press an Enter Key to read the temperature sensor 30 degrees (if the enter key is pressed)

Press an Enter Key to read the temperature sensor ...

3.Write an ECP, to receive a string using UART0 interrupt, Reverse the Content of a given string & print its result back on the serial terminal.

On the serial terminal

Enter string: VECTOR

Result: ROTCEV

Enter string: ...

4.Write an ECP, to scan a number from the serial terminal & print the binary of the same number back on the serial terminal.

eg: On the serial terminal

Enter number: 13

Binary: 0000 1101

Enter number:

Note: Keep your code in an infinite loop.

5.Write an ECP, to implement a basic calculator program using UART serial terminal.

Note: Use single digit & single operator in the expression. The result can be two digits. Keep your code in an infinite loop.

eg: On the serial terminal

Enter Expr: 10+20

Result= 30 (On the serial terminal)

Enter Expr:

6.Write an ECP, to toggle an AL LED connected to P0.17 upon generating EINT0 every time.

Note: Use the P0.16 pin to fire the EINT0 & use EINT0 as Active Low & the Falling Edge Triggered.

7.Write an ECP, to get the LED pattern option from the serial terminal using UART0 & display the output on the selected pattern on the 8 AL LEDs.

LED Connections: P1.16 to P1.23

Eg: On the serial terminal

Menu

a. LED blinking pattern

b. LED shifting pattern

Option... a (if 'a' is pressed then display an LED blinking pattern on the 8 AL LEDs & display the above menu on the serial terminal)